F12 4th CAR

MAC 1147
Precalculus: Algebra/Trig

Lecture 2: Section A.2
Exponents and Radicals

Interger exponents

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Properties of radicals

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Lecture 2: Section A.2
Exponents and Radicals

Integer Exponents

$2^3 = 2 \cdot 2 \cdot 2 = 8$

Def. If a is a real number and n is a positive integer, then

$$a^n = \overbrace{a \cdot a \cdot \ldots \cdot a}^{n \text{ factors of } a}$$

where a is the **base** and n is the **exponent** or **power**.

ex. $(-3)^4 =$ $\quad -3^4 = -1 \cdot 3^4 = -81$

$-3 \cdot -3 \cdot -3 \cdot -3 = 81$

$(-2)^3 =$ $\quad -2^3 = -1 \cdot 2^3$

$-2 \cdot -2 \cdot -2 = -8$

$= -1 \cdot 2 \cdot 2 \cdot 2 = -8$

NOTE: $0^0 =$ undefined

1) **Zero exponent:** If $a \neq 0$, then $a^0 = 1$ $\quad$ ex. $7^0 = 1$

2) **Negative exponents:** if $a \neq 0$ and n is a positive integer, then

$$a^{-n} = \frac{1}{a^n}$$

Properties of Exponents

Let a and b be nonzero real numbers, variables, or algebraic expressions, and let m and n be integers.

1. $a^m a^n = a^{m+n}$
2. $\dfrac{a^m}{a^n} = a^{m \cdot n}$
3. $(ab)^m = a^m \cdot b^m$
4. $(a^m)^n = a^{mn}$
5. $\left(\dfrac{a}{b}\right)^m = \dfrac{a^m}{b^m}$

even power

6. $|a^2| = |a|^2 = a^2$

NOTE:

1. $\left(\dfrac{a}{b}\right)^{-m} = \left(\dfrac{b}{a}\right)^m$
2. $\dfrac{a^{-m}}{b^{-n}} = \dfrac{b^n}{a^m}$

ex. Simplify each expression, writing answers without negative exponents.

1) $4a^2(-3a)^0(-2a)^3$

$4a^2 \cdot 1 \cdot (-2)^3 a^3 = 4a^2(-8)a^3 = \boxed{-32a^5}$

2) $\dfrac{6ab^{-4}}{a^{-3}b^2} = \dfrac{6a \cdot a^3}{b^2 \cdot b^4} = \boxed{\dfrac{6a^4}{b^6}}$

3) $(3ab^2c)^{-1}\left(\dfrac{2a^2b^{-1}}{c^3}\right)^{-2} = \dfrac{1}{3ab^2c} \cdot \left(\dfrac{c^3}{2a^2b^{-1}}\right)^2 =$

$= \dfrac{1}{3ab^2c} \cdot \dfrac{c^6}{2^2a^4b^{-2}} = \dfrac{c^6\cancel{b^2}}{12a^5\cancel{b^2}c} = \boxed{\dfrac{c^5}{12a^5}}$

Find x.

a) $x^2 = 36$

$\sqrt{x^2} = \pm\sqrt{36}$

$x = \pm 6$

b) $x = \sqrt{36}$

principle square root

$\boxed{x = 6 \text{ only}}$

Radicals

Def. Let a and b be real numbers and $n \geq 2$ be a positive integer. If $a = b^n$, then b is an **nth root of a.**

$16 = 2^4$, then 2 is the 4th root of 16.

Use of $\sqrt{\ }$

NOTE: $\sqrt[4]{16} = 2$

1. Both 2 and -2 are square roots of 4. We say the **principal square root** of 4, written as $\sqrt{4}$, is 2 (i.e. $\sqrt{4} = 2$).

$\sqrt{4} \neq -2$

positive answer only

2. The square roots of 4 are ± 2, but $\sqrt{4} \neq \pm 2$.

Def. Let a be a real number that has at least one nth root. The **principal nth root of a** is the nth root that has the same sign as a, denoted by $\sqrt[n]{a}$.

If even index, then use $|\ \ |$ to insure positive answer.

ex. $\sqrt[2]{7^2} = \boxed{7}$ $\qquad \sqrt[3]{\frac{8}{27}} = \sqrt[3]{\left(\frac{2}{3}\right)^3} = \boxed{\frac{2}{3}}$

$\sqrt{(-5)^2} = \sqrt{25} = \boxed{5}$ $\qquad \sqrt[3]{(-4)^3} = \sqrt[3]{-64} = \boxed{-4}$

$= |-5| =$

D.N.E. $= \sqrt{-5^2} = \sqrt{-25}$ $\qquad \sqrt[3]{-4^3} = \sqrt[3]{-1 \cdot 4^3} = \sqrt[3]{-64}$

$= \boxed{-4}$

NOTE: $\sqrt[2]{a^2} = |a|$ $\qquad \sqrt[3]{a^3} = a$

NOTE: $\sqrt{a} =$ D.N.E.; if $a < 0$.

in the reals

Properties of Radicals

Let a and b be real numbers, variables, or algebraic expressions, and let m and n be positive integers.

1. $\sqrt[n]{a^m} = (\sqrt[n]{a})^m$

2. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{a \cdot b}$

3. $\dfrac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\dfrac{a}{b}}$ $(b \neq 0)$

4. $\sqrt[m]{\sqrt[n]{a}} = \sqrt[m \cdot n]{a}$

5. $(\sqrt[n]{a})^n = a$ if $\sqrt[n]{a}$ exists

6. $\sqrt[n]{a^n} = |a|$ if n is even

   $\sqrt[n]{a^n} = a$ if n is odd

$\sqrt[4]{-16}$ DNE

$\sqrt[4]{-16} \neq -16$

if n is even, a is neg, then DNE

Simplifying Radicals

An expression involving radicals is in **simplest form** if

1. All possible factors have been removed from the radical.

2. All fractions have radical-free denominators.

3. The index of the radical is reduced.

ex. Simplify each expression.

1) $\sqrt[3]{24a^4} = \sqrt[3]{\cancel{2^3}\cdot 3\cdot \cancel{a^3}\cdot a} = \boxed{2a\sqrt[3]{3a}}$

2) $\sqrt[4]{a^8b^4c^5} = \sqrt[4]{\cancel{(a^2)^4}\,\cancel{b^4}\,\cancel{c^4}\cdot c} = \boxed{a^2|bc|\sqrt[4]{c}}$

NOTE: $\sqrt{a+b} \neq \sqrt{a} + \sqrt{b}$ $\quad \sqrt{4+9} \neq \sqrt{4} + \sqrt{9}$

$\sqrt{a^2+b^2} \neq a+b$ $\quad \sqrt{13} \neq 2+3 = 5$

Combining Radicals

We can add or subtract **like radicals** - radicals with the same index and radicand.

ex. $\sqrt{32} + \sqrt{200} = \sqrt{4^2\cdot 2} + \sqrt{10^2\cdot 2} =$

$4\sqrt{2} + 10\sqrt{2} = \sqrt{2}(4+10)$

$= \boxed{14\sqrt{2}}$

$= \sqrt{2}\cdot 14$

ex. If $a > 0$, $\sqrt{64a^7} + \sqrt{a^3}$

$\sqrt{8^2(a^3)^2 a} + \sqrt{a^2\cdot a} =$

$8a^3\sqrt{a} + a\sqrt{a} = \boxed{\sqrt{a}(8a^3+a)} = \boxed{(8a^3+a)\sqrt{a}}$

Rationalizing Denominators: multiply both numerator and denominator by an expression that allows you to eliminate radicals in the denominator.

ex. $\frac{3}{2\sqrt{6}} \cdot \frac{\sqrt{6}}{\sqrt{6}} = \frac{3\sqrt{6}}{2\sqrt{36}} = \frac{\cancel{3}\sqrt{6}}{2 \cdot \cancel{6}\,2} = \boxed{\frac{\sqrt{6}}{4}}$

ex. $\frac{1}{\sqrt[3]{2x^5}} \cdot \frac{\sqrt[3]{2^2 x}}{\sqrt[3]{2^2 x}} = \frac{\sqrt[3]{4x}}{\sqrt[3]{2^3 (x^2)^3}} = \boxed{\frac{\sqrt[3]{4x}}{2x^2}}$

Multiply by the conjugate.

ex. $\left(\frac{3}{1+\sqrt{6}}\right)\left(\frac{1-\sqrt{6}}{1-\sqrt{6}}\right) = \frac{3(1-\sqrt{6})}{1-6} = \boxed{\frac{3-3\sqrt{6}}{-5}} =$

$\left(\frac{-1}{-1}\right)\left(\frac{3-3\sqrt{6}}{-5}\right) = \boxed{\frac{-3+3\sqrt{6}}{5}} = \boxed{\frac{3\sqrt{6}-3}{5}}$

Rational Exponents

Def. If a is a real number and $n \geq 2$ is an integer, then

$$a^{1/n} = \sqrt[n]{a}$$

if $\sqrt[n]{a}$ exists. watchout for even index and negative radicand.

NOTE: If n is even and $a < 0$, then $a^{1/n}$ does not exist.

ex. 1) $81^{1/4} = \boxed{3}$

$\sqrt[4]{81} = 3$

2) $(-27)^{1/3} = \boxed{-3}$

$\sqrt[3]{-27} =$

3) $-64^{1/2} =$

$-1 \cdot 64^{1/2} = -1 \cdot 8 = \boxed{-8}$

$-1 \cdot \sqrt{64} =$

4) $(-64)^{1/2} = \sqrt{-64} =$ D.N.E.

Def. If a is a real number and m and n are positive integers

exponent index

$$a^{m/n} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m$$

if $\sqrt[n]{a}$ exists.

even neg

Usually easiest to take index first.

<u>ex.</u> Simplify:

1) $16^{3/2} = \left(\sqrt{16}\right)^3 = 4^3 = \boxed{64}$

2) $\left(-\frac{125}{8}\right)^{-2/3} = \left(\frac{-8}{125}\right)^{2/3} = \left(\sqrt[3]{\frac{-8}{125}}\right)^2$

$= \left(\frac{-2}{5}\right)^2 = \boxed{\frac{4}{25}}$

3) $\sqrt[3]{\sqrt[2]{27}} = \boxed{\sqrt[6]{27}}$

Practice.

Simplify each expression. Assume all variables are positive.

1) $\left(\dfrac{16x^{-10}}{x^6}\right)^{-1/4}$

2) $\dfrac{x^{4/3}y^{-2/3}}{(x^2y)^{1/3}}$

3) $(3-\sqrt{2})^{\sqrt{3}+1}\cdot(3+\sqrt{2})^{\sqrt{3}+1}\cdot 7^{1-\sqrt{3}}$

4) $\sqrt{\sqrt[3]{\sqrt[3]{2^6}}}$

5) $3x\sqrt{5x}+4\sqrt{20x^3}-x\sqrt{125x}$

(3) Hint: $x^3 \cdot y^3 = (xy)^3$

$(x-2)^3(x+2)^3 = [(x-2)(x+2)]^3$

$= [x^2-4]^3$

Answer. 1) $\dfrac{x^4}{2}$ 2) $\dfrac{x^{2/3}}{y}$ 3) 49 4) $\sqrt[3]{2}$ 5) $6x\sqrt{5x}$